Rayat Shikshan Sanstha's

Karmaveer Bhaurao Patil Polytechnic, Satara

Micro-Project Report

On

"Color Picker Tool"

Presented By

ROLL NO.	NAME OF THE STUDENT
126	**Mo Abrar Shoukat Shaikh**

Program: **Diploma in Computer Engineering**

Class**: Third Year (Semester VI)**

Course: **Client-Side Scripting (Subject Code: 316005)**

Guided By

Mr. Shinde A. D.

Computer Engineering Department

[2025-26]

Rayat Shikshan Sanstha's

Karmaveer Bhaurao Patil Polytechnic, Satara

CERTIFICATE

This is to certify that **Mr Mo Abrar Shoukat Shaikh** of **Third Year (Semester VI)** have successfully completed the Micro-Project work entitled "Color Picker Tool**"** in the **Client-Side Scripting (316005)** of Program **Diploma in Computer Engineering** of Maharashtra State of Technical Education, Mumbai, Maharashtra State.

Mr. Shinde A. D.	Mr. Ghorpade B. S.	Mr. Kadam B. V.
Guide	Head of Department	I/C Principal

Date:
Place:

UNDERTAKING

I, **Mo Abrar Shoukat Shaikh**, Enrollment Number **24210940248**, studying in **3rd Year, Computer Engineering** Branch, **VI** Semester **2025-2026** Year, hereby undertake that I will preserve the micro-project and its report in my custody until the completion of my program. I assure that I will produce the same without fail whenever I am asked to do so.

Date: _____

Place: _____

Signature of Student: _____

Annexure – I A Part A – Plan

Micro-Project Proposal

Color Picker Tool

1.0 Brief Introduction:

The Color Picker Tool is a web-based application that allows users to select and preview colors in real-time. It supports HEX and RGB formats and provides features like copy-to-clipboard and color history tracking. This project is important for web designers and developers to quickly choose and manage colors efficiently.

2.0 Aim of the Micro-Project:

This micro-project aims at developing an interactive color selection tool using JavaScript that improves user experience and simplifies color management.

a. To implement real-time color selection

b. To display HEX and RGB values dynamically

c. To provide clipboard and history features

3.0 Action Plan

Sr. No.	Details of Activity	Planned Start date	Planned Finish date	Name of Responsible Team Members
1.	**Requirement study & planning** (guided by Mr. Shinde A.D.)			
2.	**Design of interface & modules**.			
3.	**HTML structure development**			
4.	**CSS styling & responsiveness**			
5.	**JavaScript implementation**			
6.	**Integration of components**			
7.	**Testing & validation**			
8.	**Documentation & submission**			

4.0 Resources Required and Used

Sr. No.	Name of resource / material	Specification	Quantity	Remarks
1	Hardware	Computer/ Laptop: Minimum i3/i5, 4GB RAM	1	
2	Software	Platform - Windows 10,11. Software tools – VS Code, Browser, GitHub. MS Word.	1	

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Chapter			Contents	Page No.
1.			Introduction	1-5
2.			Diagrams	
	2.1		Flowchart	5-10
	2.2		Architecture Diagram	11-15
	2.3		Block Diagram	16-20
	2.4		Class Diagram	21-25
3.			Source Code	25-30
4.			Output (Screenshot)	30-35
5.			Conclusion	35-40
6.			References	45-50

CONTENTS

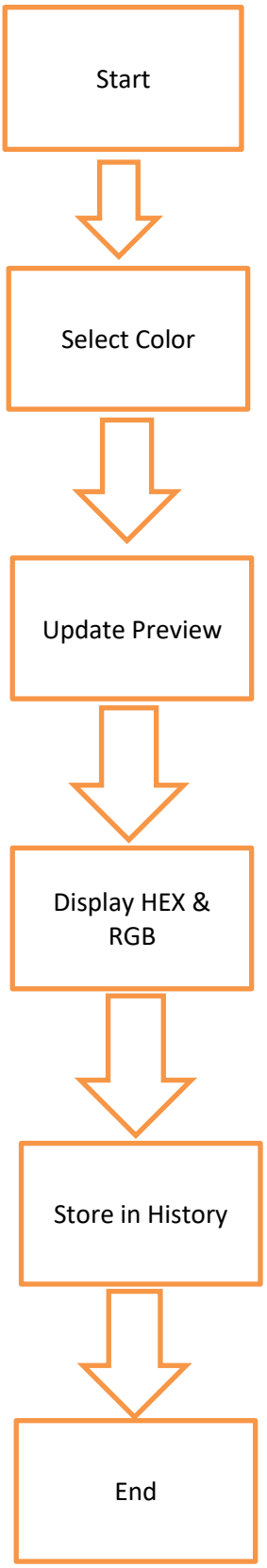
CHAPTER 1: INTRODUCTION

The Color Picker Tool is designed to provide an easy and efficient way to select colors for web applications. It allows users to pick colors using a color input tool and view them in HEX and RGB formats. The tool also includes features like real-time preview, clipboard copy, and color history.

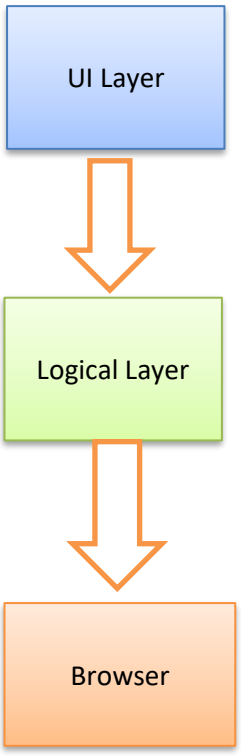
This project demonstrates the use of JavaScript for DOM manipulation, event handling, and real-time UI updates.

CHAPTER 2: DIAGRAMS

2.1 Flowchart:



2.2 Architecture Diagram:



2.3 Block Diagram.

2.4 Class Diagram:

CHAPTER 3: SOURCE CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Color Picker Tool</title>
<style>
 Body
 {
          font-family: Arial, sans-serif;
          background: #f4f4f4;
          display: flex;
          justify-content: center;
         padding: 20px;
 }

 .container
 {
          background: #fff;
         padding: 20px;
        border-radius: 12px;
        width: 320px;
         box-shadow: 0 5px 15px rgba(0,0,0,0.1);
 }

 h2
 {
         text-align: center;
 }
```

```
.preview
{
        height: 100px;
        border-radius: 10px;
         margin: 15px 0;
        border: 1px solid #ccc;
}

input, button
{
        width: 100%;
       padding: 8px;
        margin: 6px 0;
        border-radius: 8px;
       border: 1px solid #ccc;
}

button
{
       cursor: pointer;
       background: #333;
        color: white;
}

.history
{
       display: flex;
        flex-wrap: wrap;
       gap: 5px;
       margin-top: 10px;
}
```

```
 .swatch
{
        width: 25px;
        height: 25px;
        border-radius: 5px;
        cursor: pointer;
        border: 1px solid #ddd;
 }


 .error

{
        color: red;
        font-size: 12px;
 }

 .toast {
        position: fixed;
        bottom: 20px;
        right: 20px;
        background: black;
        color: white;
        padding: 8px 12px;
        border-radius: 6px;
        display: none;
 }

</style>
</head>

<body>
```

```
<div class="container">
        <h2>Color Picker Tool</h2>

  <!-- Color Input -->

        <input type="color" id="colorPicker">

  <!-- Preview -->

        <div class="preview" id="preview"></div>

  <!-- HEX -->

         <input type="text" id="hexInput" placeholder="#FFFFFF">
        <button onclick="copyText('hexInput')">Copy HEX</button>

  <!-- RGB -->

        <input type="text" id="rgbInput" placeholder="rgb(255,255,255)">
         <button onclick="copyText('rgbInput')">Copy RGB</button>

  <div class="error" id="error"></div>

  <!-- History -->

        <div class="history" id="history"></div>
         <button onclick="clearHistory()">Clear History</button>
</div>

<div class="toast" id="toast">Copied!</div>

<script>
        const picker = document.getElementById("colorPicker");
```

```
        const preview = document.getElementById("preview");
        const hexInput = document.getElementById("hexInput");
        const rgbInput = document.getElementById("rgbInput");
        const error = document.getElementById("error");
        const historyDiv = document.getElementById("history");

let history = [];

/* HEX → RGB */

function hexToRgb(hex) {
        hex = hex.replace("#", "");
         if (hex.length === 3) {
         hex = hex.split('').map(x => x + x).join('');
  }

        const bigint = parseInt(hex, 16);
        if (isNaN(bigint)) return null;

                return {
         r: (bigint >> 16) & 255,
        g: (bigint >> 8) & 255,
        b: bigint & 255
  };
}

/* RGB → HEX */

        function rgbToHex(r, g, b) {
                return "#" + [r, g, b]
                 .map(x => x.toString(16).padStart(2, '0'))
                 .join('')
                  .toUpperCase();
```

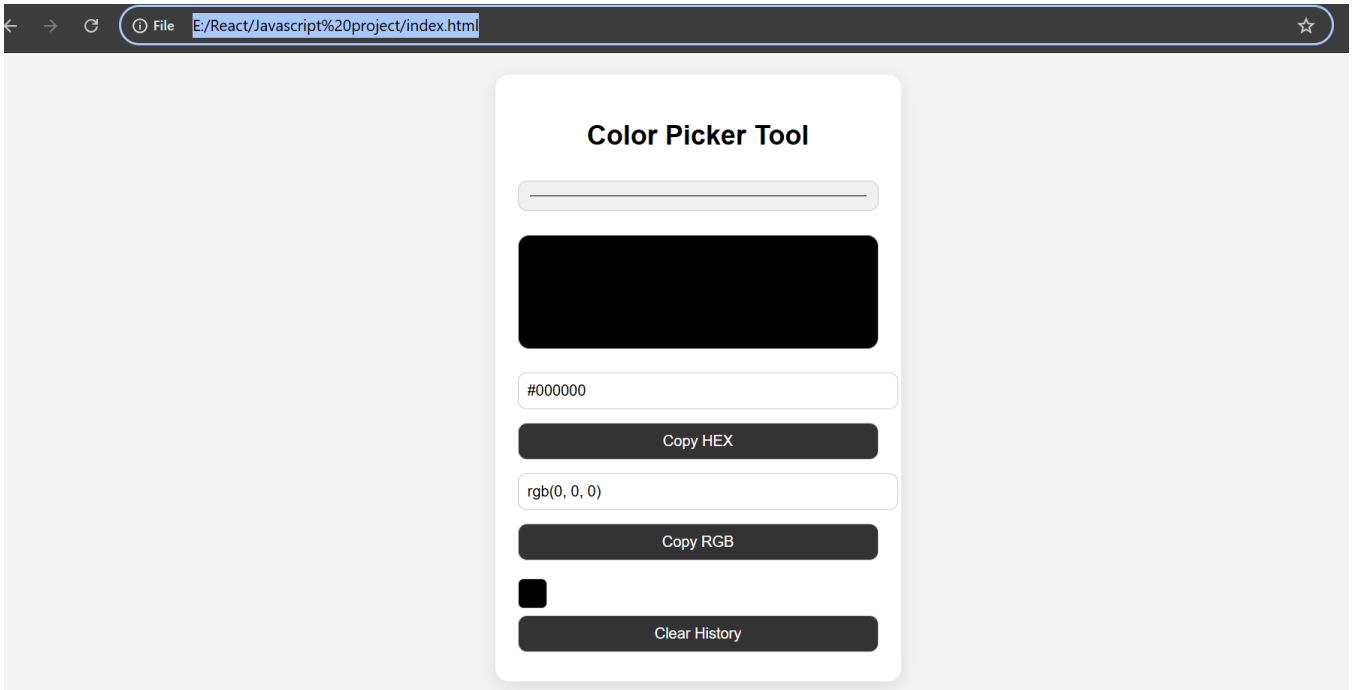
```
}

/* Update UI */

        function updateColor(hex) {
          const rgb = hexToRgb(hex);
          if (!rgb) return;

          const rgbStr = `rgb(${rgb.r}, ${rgb.g}, ${rgb.b})`;

          preview.style.background = hex;
          hexInput.value = hex.toUpperCase();
          rgbInput.value = rgbStr;

          addToHistory(hex);
}

/* Picker Event */

        picker.addEventListener("input", e => {
          updateColor(e.target.value);
});

/* HEX Manual Input */

        hexInput.addEventListener("input", () => {
          let val = hexInput.value.trim();

         if (!val.startsWith("#")) val = "#" + val;

          if (/^#([0-9A-F]{3}|[0-9A-F]{6})$/i.test(val)) {
           error.textContent = "";
          updateColor(val);
```

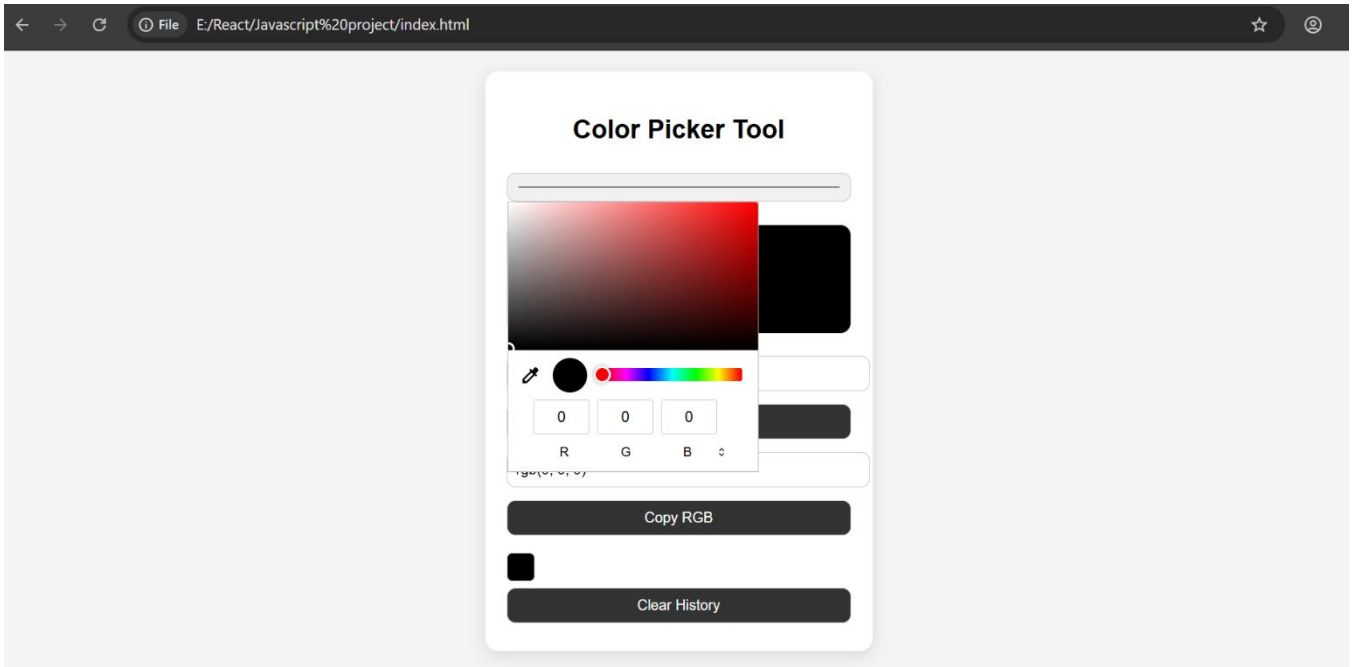
```
        } else {
         error.textContent = "Invalid HEX format";
  }
});

/* RGB Manual Input */

        rgbInput.addEventListener("input", () => {
         const match = rgbInput.value.match(/\d+/g);

         if (!match || match.length !== 3) {
          error.textContent = "Invalid RGB format";
         return;
  }

  let [r, g, b] = match.map(Number);

  if (r > 255 || g > 255 || b > 255 || r < 0 || g < 0 || b < 0) {
        error.textContent = "RGB range: 0–255";
        return;
  }

  error.textContent = "";

  const hex = rgbToHex(r, g, b);
  updateColor(hex);
});

/* Copy */

        function copyText(id) {
                const text = document.getElementById(id).value;
```

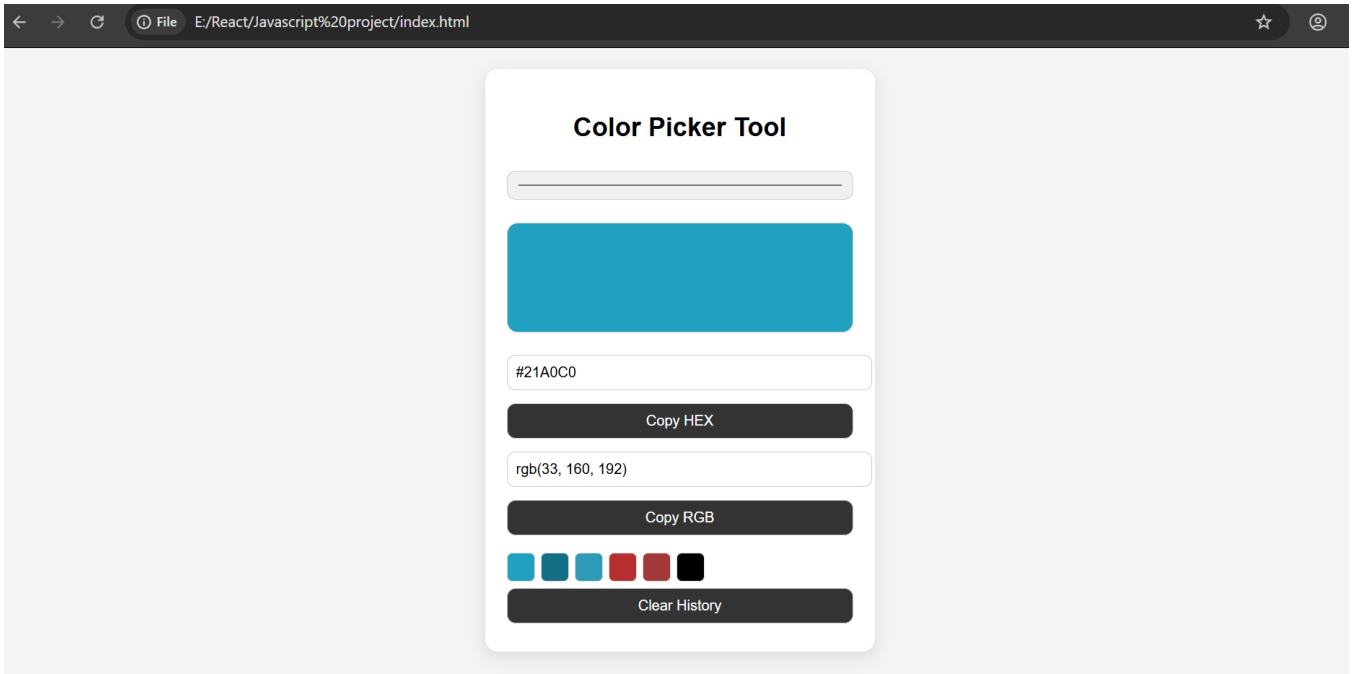
```
                if (!navigator.clipboard) {
                  alert("Clipboard not supported");
                return;
  }

          navigator.clipboard.writeText(text)
            .then(() => showToast())
          .catch(() => alert("Copy failed"));
}

/* Toast */

        function showToast() {
                const toast = document.getElementById("toast");
                 toast.style.display = "block";
                setTimeout(() => toast.style.display = "none", 1500);
}

/* History */

        function addToHistory(color) {
                if (history.includes(color)) return;

                history.unshift(color);
                if (history.length > 20) history.pop();

                renderHistory();
}

function renderHistory() {
        historyDiv.innerHTML = "";
         history.forEach(color => {
         const swatch = document.createElement("div");
```

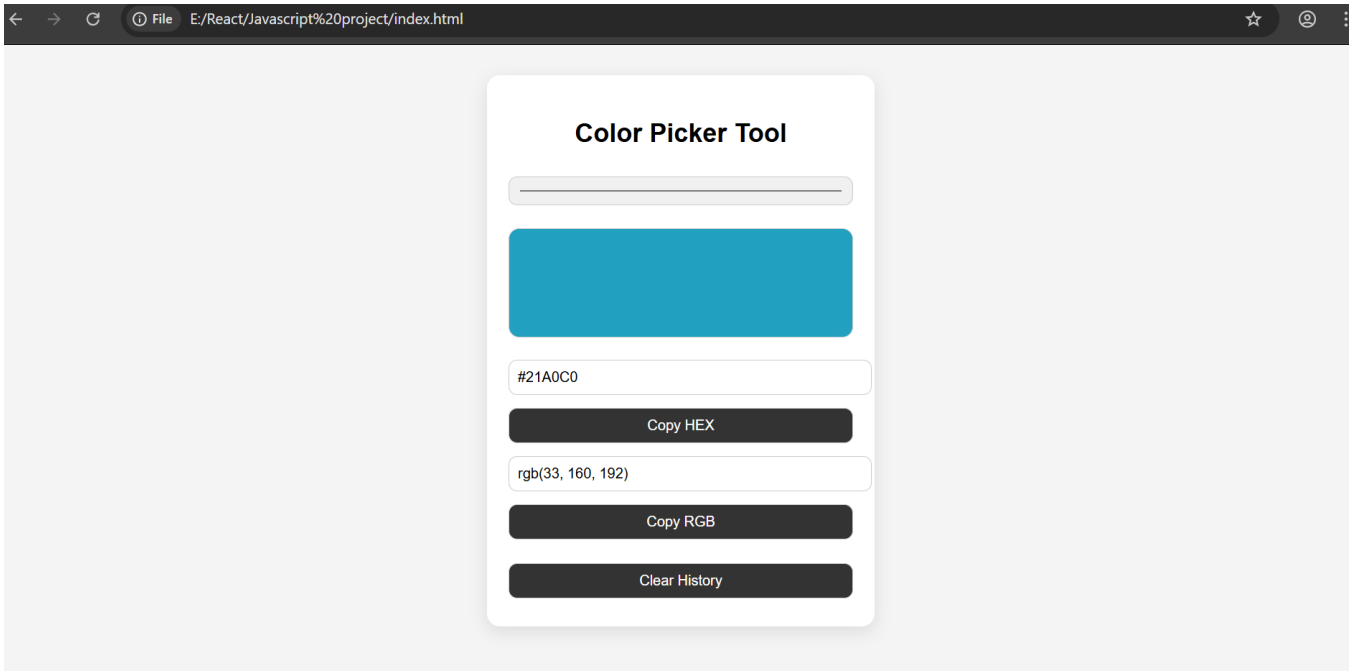
```
        swatch.className = "swatch";
         swatch.style.background = color;
        swatch.onclick = () => updateColor(color);
                historyDiv.appendChild(swatch);
  });
}

function clearHistory() {
        history = [];
        renderHistory();
}

/* Default Load */

updateColor("#000000");
</script>

</body>
</html>
```

CHAPTER 4: OUTPUT (SCREENSHOT)






File
E:/React/Javascript%20project/index.html
Color Picker Tool
#21A0C0
Copy HEX
rgb(33, 160, 192)
Copy RGB
Clear History


File
E:/React/Javascript%20project/index.html
Color Picker Tool
#21A0C0
Copy HEX
rgb(33, 160, 192)
Copy RGB
Clear History

CHAPTER 5: CONCLUSION

The Color Picker Tool was successfully developed using HTML, CSS, and JavaScript. It meets all functional requirements such as real-time color selection, format conversion, and user interaction features.

The project helped in understanding DOM manipulation, event handling, and UI design principles.

CHAPTER 6: REFERENCES

- W3Schools – JavaScript & HTML
- MDN Web Docs
- JavaScript.info
- CSS Tricks

Annexure – III Teacher Evaluation Sheet

Name of Student: Mo Abrar Shoukatx` Shaikh **Enrollment No.:** 24210940248

Name of Program: Computer Engineering **Semester:** CO-6K

Course Title: Client-Side Scripting (CSS)

Code: 316005

Title of the Micro-Project: Color Picker Tool

Course Outcomes Achieved: Understand how entrepreneurs pitch ideas, how investors evaluate businesses, and gain insights into funding, negotiation, and startup strategies.

Evaluation as per suggested Rubric for Assessment of Micro-Project

(Please tick in appropriate cell for each characteristic)

Sr. No.	Characteristic to be assessed	Poor	Average	Good	Excellent
1	Relevance to the Course Curriculum				
2	Literature Review and Information Collection				
3	Clarity of Project Proposal and Planning				
4	Achievement of Objectives as per Proposal				
5	Analysis of Data and Output Representation				
6	Quality and Functionality of Prototype / Application				
7	Report Writing and Documentation				
8	Level of Understanding during Viva / Oral Examination				

Micro-Project Evaluation Sheet

Process assessment		Product assessment		Total Marks 25
Part A Project Proposal (5 marks)	Project Methodology (5 marks)	Part B Project Report / Working Model (5 marks)	Individual Presentation / Viva (10 marks)	

Comments / suggestion about teamwork / leadership / inter-personal communication (if any):

_____

_____

_____

_____

Any other Comment:

_____

_____

_____

_____

Name and designation of faculty member: _____

Signature: _____